

Algebra general exam. January 8, 2020, 9am-1pm

Your UVa ID Number:

Directions.

- Please show all your work and justify any statements that you make.
 - State clearly and fully any theorem you use.
 - Vague statements and hand-waving arguments will not be appreciated.
 - You may assume the statement in an earlier part proven in order to do a later part.
- DO EACH PROBLEM ON A SEPARATE SHEET OF PAPER, AND STAPLE THEM TOGETHER IN THE CORRECT ORDER BEFORE TURNING THE EXAM IN.

Sign below the pledge:

"On my honor, I pledge that I have neither given nor received help on this assignment."

Problem 1

- (a) (7 pts) Let C be a cyclic group of order $n \geq 2$. Explain briefly why $\text{Aut}(C) \cong \mathbb{Z}_n^\times$ (the multiplicative group of $\mathbb{Z}_n$) and why $\text{Aut}(C) \cong \mathbb{Z}_{n-1}$ if n is prime.
- (b) (7 pts) Let G be a finite group, let p be the smallest prime dividing $|G|$, and suppose that G contains a normal subgroup C of order p . Prove that C lies in the center of G .

Problem 2

(10 pts) Let A be the subgroup of $\mathbb{Z}^2$ generated by $(2, 6)$ and $(4, 8)$. Prove that $A \cong \mathbb{Z}^2$ and find (with proof) elements $v_1, v_2 \in \mathbb{Z}^2$ and $n_1, n_2 \in \mathbb{N}$ such that $\mathbb{Z}^2 = \mathbb{Z}v_1 \oplus \mathbb{Z}v_2$ and $A = \mathbb{Z}(n_1v_1) \oplus \mathbb{Z}(n_2v_2)$.

Problem 3

(10 pts) Let $R = \mathbb{Z}[i]$, the ring of Gaussian integers. Find (with complete proof!) the number of ideals of R which contain 30.

Problem 4

Let R be a commutative ring with 1 and let $x \in R$.

- (a) (3 pts) Prove that $x \notin R^\times$ if and only if $x \in M$ for some maximal ideal M of R .
- (b) (9 pts) Prove that the following are equivalent:
 - (i) $x \in M$ for every maximal ideal M of R
 - (ii) $1 + xy \in R^\times$ for every $y \in R$

Problem 5

(12 pts) Let F be a field and let $A, B \in \text{Mat}_n(F)$ for some $n \in \mathbb{N}$. Suppose that $A^2 = B^2 = I$ and $\text{rk}(A - I) = \text{rk}(B - I)$. Prove that A and B are similar. Hint: Consider separately the cases $\text{char } F \neq 2$ and $\text{char } F = 2$.

Problem 6

In each part of this problem determine if the given objects are isomorphic:

- (a) (4pts) $\mathbb{R}$ and $\mathbb{C}$ as $\mathbb{Q}$ -vector spaces
- (b) (3pts) $\mathbb{C}$ and $\mathbb{R} \times \mathbb{R}$ as rings
- (c) (4 pts) $\mathbb{R}[x] / (x^2 - 1)$ and $\mathbb{R} \times \mathbb{R}$ as rings
- (d) (4 pts) $\mathbb{R}[x] / (x - 1)$ and $\mathbb{R}[x] / (x + 1)$ as $\mathbb{R}[x]$ -modules

Problem 7

Let F, K_1, K_2 and L be fields with $F \subseteq K_1 \subseteq L$ and $F \subseteq K_2 \subseteq L$. Recall that the tensor product $K_1 \otimes_F K_2$ has a natural structure of a ring where multiplication of simple tensors is given by

$$(a \otimes b) \cdot (c \otimes d) = ac \otimes bd \text{ for all } a, c \in K_1 \text{ and } b, d \in K_2.$$

- (a) (5 pts) Prove that there exists a unique ring homomorphism $\pi : K_1 \otimes_F K_2 \rightarrow L$ such that $\pi(a \otimes b) = ab$ for all $a \in K_1, b \in K_2$.
- (b) (2 pts) Assume that $K_1 \otimes_F K_2$ is a field. Prove that π is injective.
- (c) (5 pts) Assume that $K_1 \cap K_2 \neq F$. Prove that π is not injective.

Problem 8

Let K/F be a field extension. Suppose that $K = F(a, b)$ for some $a, b \in K$ such that $a^2 \in F$ and $b^2 \in F$.

- (a) (3 pts) Prove that $[K : F] \leq 4$
- (b) (4 pts) Give a specific example (with full proof) where $[K : F] = 4$

(c) (4 pts) Assume that $\text{char}(F) \neq 2$. Prove that the extension K/F is Galois

(d) (4 pts) Now assume that F is finite. Prove that $[K : F] \leq 2$.